Federico Tesone

Principal Engineer · Search, Retrieval & Agentic AI Systems

New York, NY · US Permanent Resident | fedetesone.github.io | linkedin.com/in/ftesone

Summary

Principal engineer with 14+ years of software engineering experience, specializing in search, retrieval, and agentic AI systems. Led Daydream's 9-person Search Core team, which re-architected conversational search from a monolithic Elasticsearch engine into a workflow-based pipeline with LLM orchestration, preference memory, and evaluation-gated releases. Previously set technical strategy at Grubhub across 3 teams and 20 engineers.

Experience

Principal Engineer, Daydream – New York, NY, USA *July 2025 – Oct 2026*

Led a 9-person Search Core team owning retrieval, ranking, and query understanding for an AI-native fashion search and conversational engine.

- Re-architected a monolithic search system, which ran hybrid retrieval inside Elasticsearch, into a workflow-based pipeline that combines reciprocal rank fusion (RRF) across text, visual, and semantic retrieval over 5M+ product options, cross-encoder re-ranking, and a feedback loop; extended it to image search and visual discovery.
- Converted query understanding from a single blocking LLM call into a concurrent DAG with confidence routing and multi-intent splitting, reducing its P95 latency by 64%, from 7s to 2.5s, while extracting substantially more signal per query.
- Designed multi-pass re-ranking with agentic fallback loops and added a planner/judge stage that re-ranks retrieved products using structured decision models, raising the LLM-judged quality score from 0.66 to 0.75 and increasing the merchant click-out rate by 12% in an A/B test.
- Built the evaluation and observability foundation: offline and online evaluation that gates ranking and prompt changes, and end-to-end tracing of LLM decisions, retrieval stages, ranking signals, latency, and final results.
- Owned the runtime agentic pipeline, its offline jobs, and the user-insights service (Java, Python) in production, including multi-turn preference memory and embedding pipelines for personalization.

Staff Engineer, Grubhub – New York, NY, USA *Feb 2023 – July 2025*

Technical lead of an 8-engineer search team; set technical strategy and architecture across Search, Home Page, and Data Science (3 teams, 20 engineers).

- Architected a modular search runtime separating context, candidate selection, ranking, experimentation, and hydration, enabling independent evolution of retrieval and ML ranking systems.
- Owned Grubhub's search service, ML model-serving service, and data pipelines (Java, Scala/Spark, Python) in production, serving approximately 120K–150K requests per minute across major consumer surfaces.
- Led the work that halved end-to-end search P95 latency, from about 1s to 500ms.
- Introduced multi-query retrieval and result blending, improving search conversion by 2.37%, then redesigned it as a single-query architecture that preserved the gain while reducing Elasticsearch CPU utilization by 34% and saving $134K in annual compute costs.
- Designed candidate selection and ranking infrastructure integrating real-time ML model predictions, embeddings, business constraints, geographic filtering, and experimentation signals.
- Mentored senior engineers across 3 teams.

Senior Software Engineer, Grubhub – New York, NY, USA *Dec 2019 – Feb 2023*

Engineered high-availability Java microservices for the discovery runtime platform.

- Built a low-latency vector retrieval system for production runtime workloads using Annoy, with custom sharding of vector indexes to support scalable embedding-based lookup across large datasets.
- Developed low-latency data serving infrastructure for embedding-based ML models and user targeting.
- Partnered with machine-learning and data teams to productionize new ranking and recommendation capabilities.

Senior Developer, Endava – Rosario, Argentina *July 2016 – Dec 2019*

Developed backend systems and microservices using Java, Scala, Python, Cassandra, AWS, and CI/CD pipelines.

Auxiliary Professor, Universidad Abierta Interamericana – Rosario, Argentina *Aug 2014 – Mar 2019*

Taught Object-Oriented Software Design, GoF Design Patterns, Software Metrics, QA, and Automated Testing.

Earlier Experience

Team Lead & Senior Developer, Nación Servicios (2013–2016): Core banking loyalty platforms, transaction authorization systems, and EMV payment services.

BI Consultant, BPD Solutions (2012–2013): Enterprise ETL pipelines, data integration workflows, and data warehousing systems.

Education

Universidad Tecnológica Nacional, MSc in Software Engineering *2013 – Nov 2021*

Thesis: Developing a Similarity Measure for Recommender Systems on Community Question Answering Sites: A Big Data Analysis Using a Clustering Ensemble Method. Written in Spanish: *Desarrollo de una medida de similaridad para Sistemas de Recomendación en sitios de Community Question Answering. Análisis desde un enfoque Big Data y usando un método de ensamble de clustering.*

Universidad Tecnológica Nacional, BSc in Information Systems Engineering *2007 – 2012*

5-year degree with minor thesis.

Technical Expertise

AI & Search Systems: Hybrid Retrieval, Dense & Sparse Retrieval, Vector Search, Ranking & Re-ranking, LLM Query Understanding, Agentic Search, Multi-turn Preference Memory, Evaluation Systems, Embeddings, Personalization, Recommendation Systems

Backend & Data Infra: High-Throughput, Low-Latency Services, Real-Time Data Serving, Feature Infrastructure, Event-Driven Systems, Observability, Reliability Engineering

Languages & Tech: Java (primary), Python, Scala, Spark, SQL, Elasticsearch, Cassandra, ScyllaDB, Redis, Google Cloud, AWS

Languages: English (Full Professional), Spanish (Native)

Honors and Awards

Cuadro de Honor - Honor Roll

Published in the list of highest grade point averages of the Province of Santa Fe, Argentina.